

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

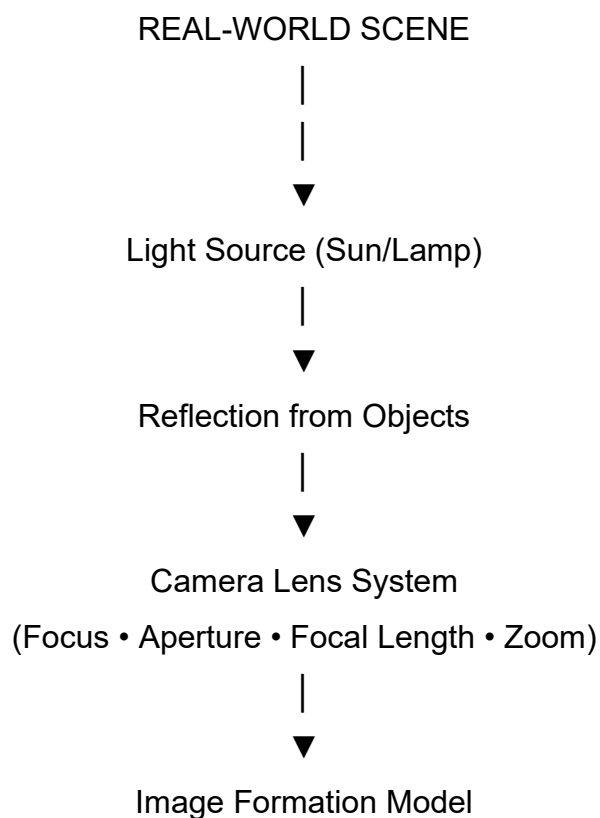
CO1: *Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)*

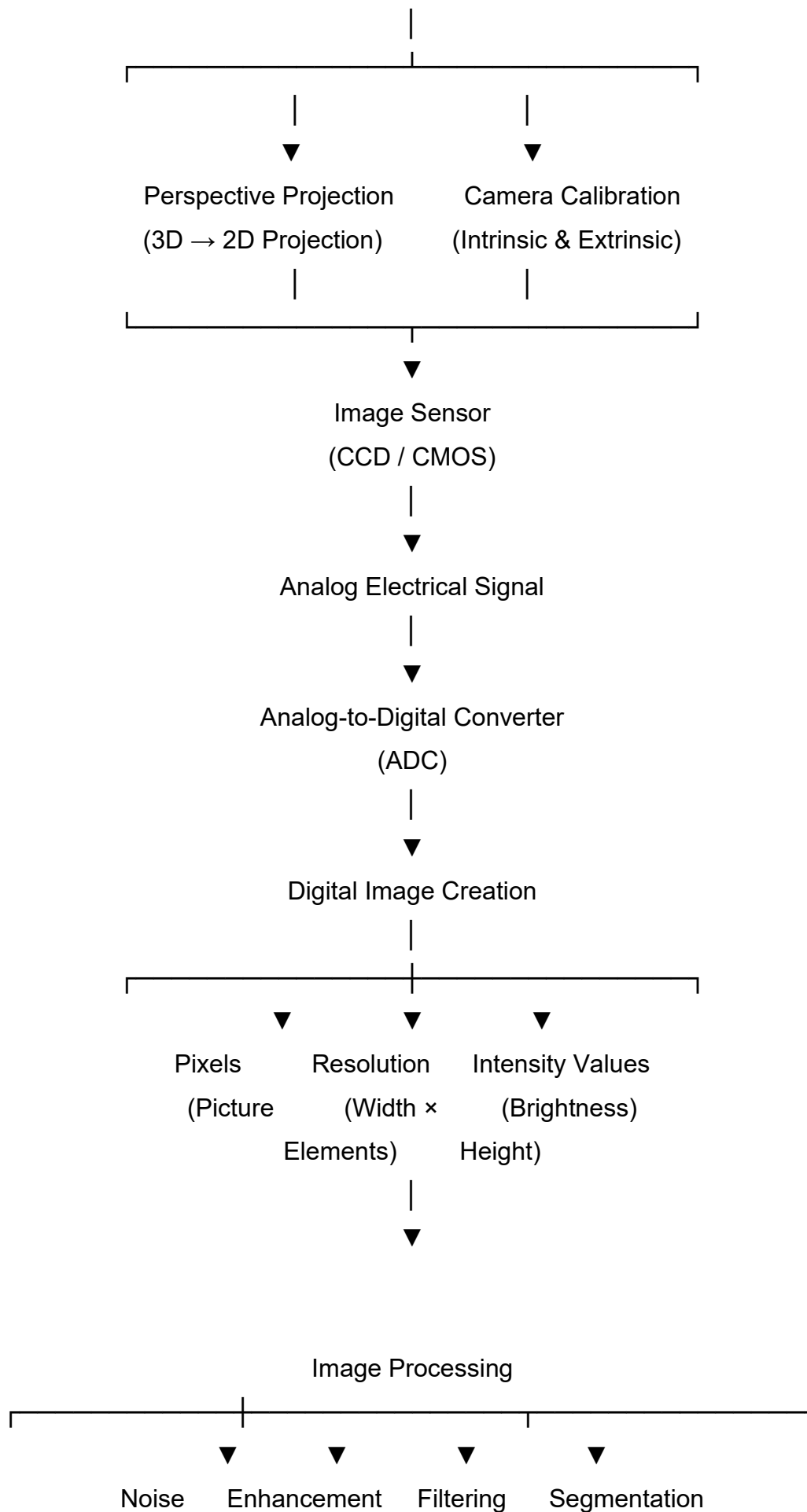
Assessment Tool Weightage: 20%

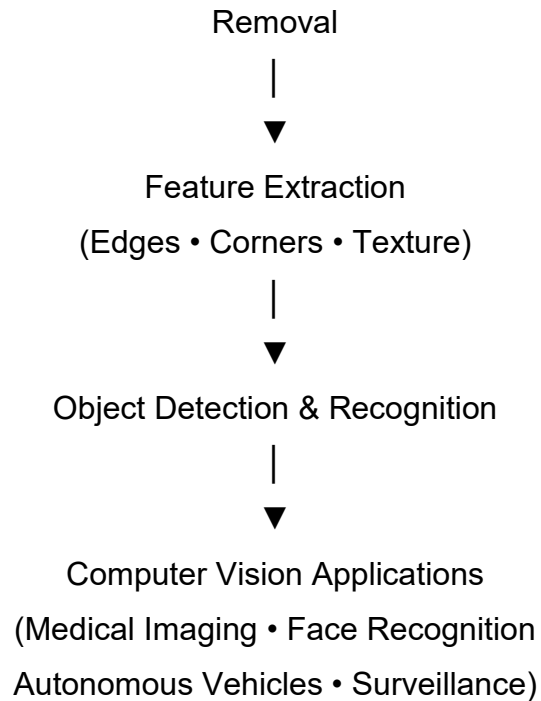
1. Construct a comprehensive and well-organized concept map that accurately identifies all key concepts from literature involved in the pipeline from real-world scene capture to digital image representation, establishes clear and meaningful relationships among intermediate transformations and dependencies, integrates insights from multiple studies, and presents the information in a clear and visually structured manner.

Question 1

Concept Map: From Real-World Scene to Digital Image Representation





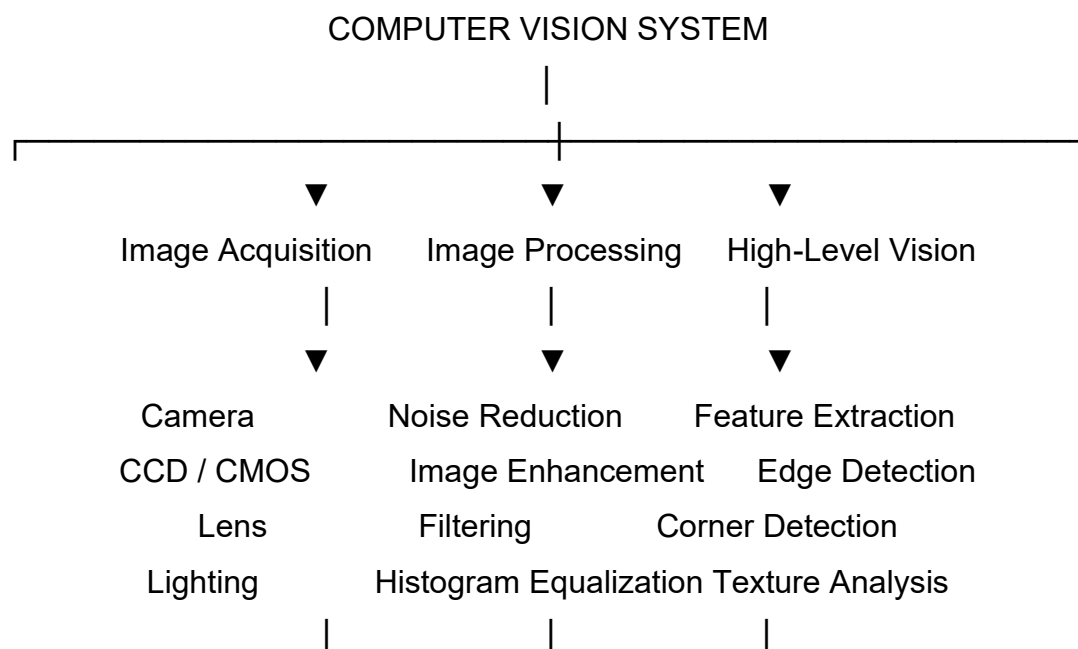


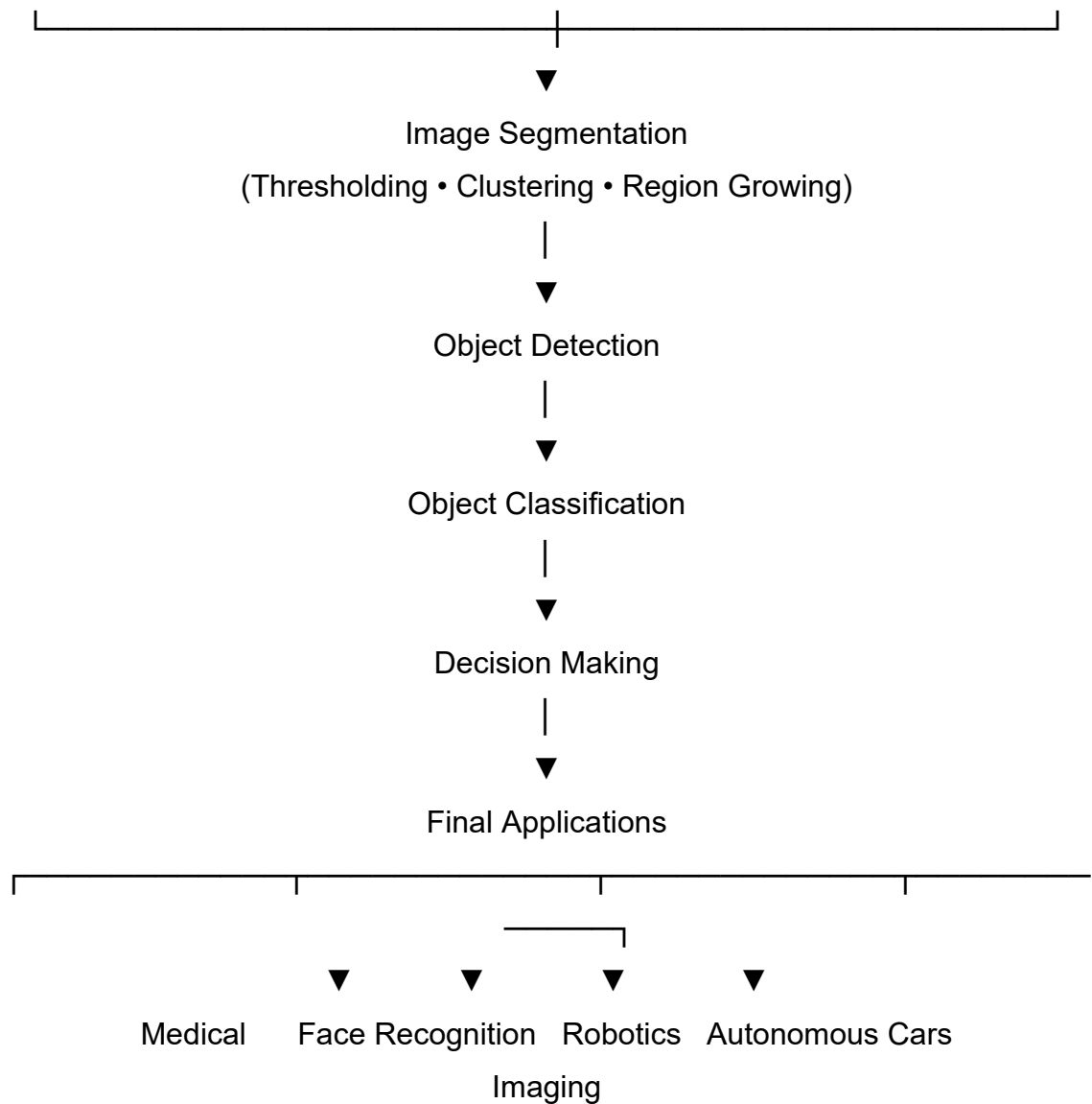
Literature Integration

- **Richard Szeliski (Computer Vision: Algorithms and Applications, 2022)** – Image formation and camera models.
- **Gonzalez & Woods (Digital Image Processing, 2018)** – Image acquisition and preprocessing.
- **Forsyth & Ponce (Computer Vision: A Modern Approach, 2012)** – Feature extraction and recognition.

Question 2

Concept Map: Major Stages of Visual Data Handling in Computer Vision



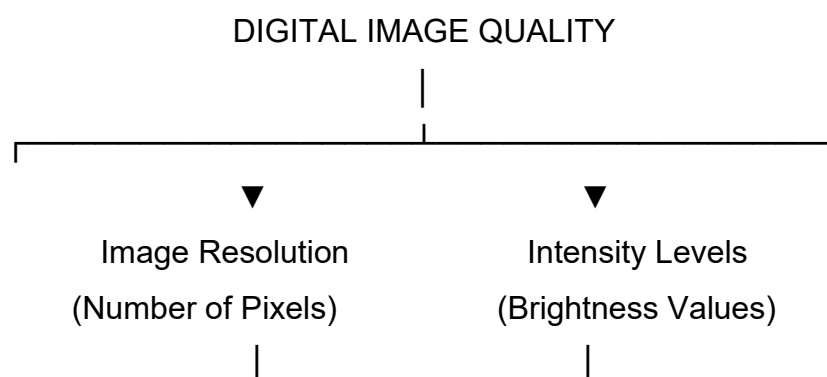


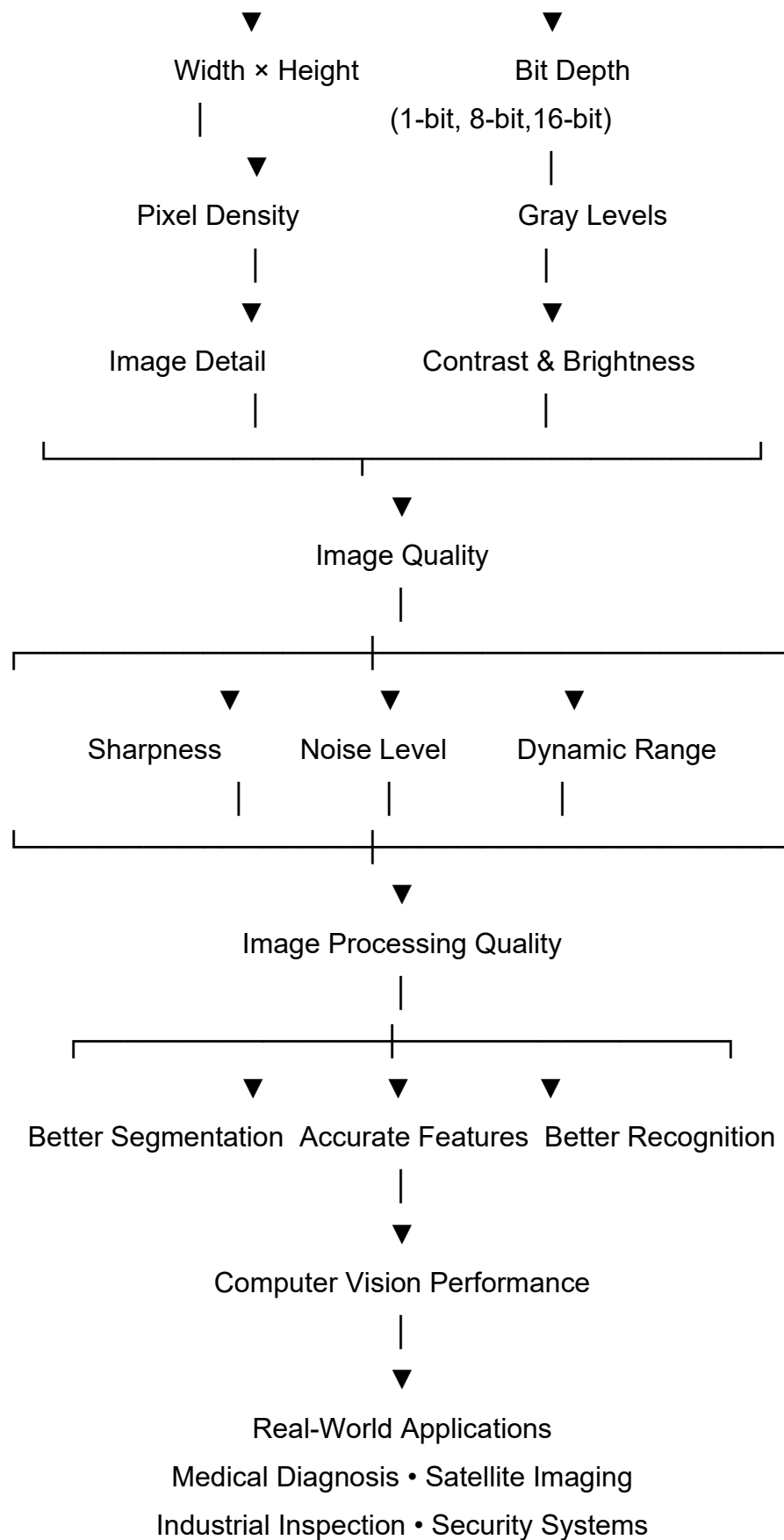
Literature Integration

- **Szeliski (2022)** – Complete Computer Vision pipeline.
- **Forsyth & Ponce (2012)** – Segmentation and feature extraction.
- **Gonzalez & Woods (2018)** – Image enhancement techniques.

Question 3

Concept Map: Image Resolution and Intensity Levels





Cause–Effect Relationships

Cause	Effect
Higher Resolution	More image details

Higher Bit Depth	Better intensity representation
Better Contrast	Easier feature extraction
Less Noise	Improved segmentation
Better Image Quality	Higher recognition accuracy

Literature Integration

- **Gonzalez & Woods (Digital Image Processing, 2018)** – Resolution, intensity levels, and image quality.
- **Szeliski (2022)** – Image representation and computer vision fundamentals.
- **Forsyth & Ponce (2012)** – Importance of image quality in recognition systems.

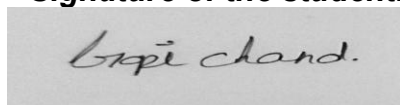
References (IEEE Format)

1. R. Szeliski, *Computer Vision: Algorithms and Applications*, 2nd ed., Springer, 2022.
2. R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed., Pearson, 2018.
3. D. A. Forsyth and J. Ponce, *Computer Vision: A Modern Approach*, 2nd ed., Pearson, 2012.

Code of Conduct:

*I **G. Gopi Chand (192424384)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately	Identifies most key concepts with minor	Identifies limited concepts; some are	Fails to identify essential concepts

		and comprehensively	omissions	irrelevant or missing	
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration ; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand